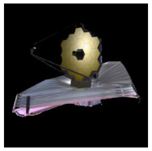

9138 - Investigating the Cumulative Feedback of High-Redshift Restarted Radio AGN on Massive Galaxy Quenching

Cycle: 4, Proposal Category: GO

INVESTIGATORS

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OBSERVATIONS

Folder	*Observation*	*Label*	*Observing Template*	*Science Target*
1 - J012230				
	1	J012230.18+270832.3	NIRSpec Fixed Slit Spectroscopy	(1) J012230.18+270832.3
2 - J095431				
	2	J095431.30+503215.4	NIRSpec Fixed Slit Spectroscopy	(2) J095431.30+503215.4
3 - J120002				
	3	J120002.30+572212.7	NIRSpec Fixed Slit Spectroscopy	(3) J120002.30+572212.7
4 - J153837				
	4	J153837.14+400640.2	NIRSpec Fixed Slit Spectroscopy	(4) J153837.14+400640.2

ABSTRACT

Quenching of massive galaxies is a significant challenge for cosmological models attempting to reproduce the observable Universe. A variety of quenching timescales are found in massive galaxies, with ~25% of them quenching in a duration of 500 Myr. AGN feedback is an important quenching mechanism, however, the small duration of an AGN phase (10-100 Myr) compared to the galaxy lifetime, makes it hard to connect these two. In this regard, restarted radio-AGN are unique, since they can be used to trace multiple epochs of AGN activity over hundreds of Myr. Restarted sources with sustained jet activity over 300-400 Myr, can suppress star formation, and quench massive galaxies over a short timescale. We propose NIRSpec fixed-slit mode observations of 4 restarted radio AGN at z~1 with massive quiescent host galaxies, to find evidence for the cumulative impact of AGN feedback. We plan to perform a stellar population modelling using spectrophotometric data, and determine the abundances of different stellar populations and the star formation history (SFH) of the galaxies. We will compare the young stellar population and recent SFH (<500 Myr) with the radio-AGN timescales, to find for the first time, evidence of cumulative AGN feedback. These observations, located at the cosmic midpoint of the Universe, will help constrain the quenching mechanisms in massive galaxies.

OBSERVING DESCRIPTION

We request NIRSpec fixed-slit mode observations of four restarted radio-AGN galaxies at z~1 with the G140M/F070 LP dispenser/filter. We aim to determine the stellar populations and star formation histories of these galaxies. We require S/N~20 at ~0.8-0.9 micron for our analysis, which includes using the Balmer lines at these wavelengths. We require 2 hrs of exposure time for three, and 1 hr exposure time for one target to achieve this. We summarise the setups below:

For targets J012230.18+270832.3, J120002.30+572212.7 and J095431.30+503215.4
Target acquisition: Same as target observation
- Acq subarray: FULL
- Acq filter: CLEAR
- Readout mode: NRSRAPID

Science observation:
- Slit aperture: S400A1
- Disperser filter: G140M/F070 LP
- Total number of ditherings: 6
  - Principal dithering pattern: 3
  - Subpixel dithering: spatial 2

- 40 groups
- 2 integrations per dither position
- Readout mode: NRSIRS2RAPID
- Subarray: FULL

--> Total exposure time on source: 2 h per target

For target J153837.14+400640.2

Target acquisition: Same as target observation

- Acq subarray: FULL
- Acq filter: CLEAR
- Readout mode: NRSRAPID

Science observation:

- Slit aperture: S400A1
- Disperser filter: G140M/F070 LP
- Total number of ditherings: 6
  - Principal dithering pattern: 3
  - Subpixel dithering: spatial 2
- 40 groups
- 1 integration per dither position
- Readout mode: NRSIRS2RAPID
- Subarray: FULL

--> Total exposure time on source: 1 h

Total requested time: 6.9 hours of science time and 12.6 hours of charged time.

Fixed Targets

#	Name	Target Coordinates	Targ. Coord. Corrections	Miscellaneous
(1)	J012230.18+270832.3	RA: 01 22 29.9581 (20.6248254d) Dec: +27 08 39.78 (27.14438d) Equinox: J2000		

Comments: Redshift = 1.144

RA (degrees) = 20.6248256082825, DEC (degrees) = 27.1443821071447
error_RA (arcsec) = 0.05170400635, error_DEC (arcsec) = 0.05368708672

Here I have provided the final target coordinates with errors. These are in the ICRS frame, and have been taken from the Tractor catalogues of Legacy Surveys DR10 and DR9. I have checked the position using the z-band images from Legacy Surveys. The astrometry of Legacy Surveys DR8 onwards is placed on the GAIA DR2 frame. The differences between the GAIA DR2 and GAIA DR3 frame are of the order of a few dozen micro-arcsec at most, and usually even much smaller. So I would think this would not matter for our case.
Category=Galaxy
Description=[Active galaxies, Radio cores, Radio jets]
Extended=YES

#	Name	Target Coordinates	Targ. Coord. Corrections	Miscellaneous
(2)	J095431.30+503215.4	RA: 09 54 30.9243 (148.6288512d) Dec: +50 32 20.10 (50.53892d) Equinox: J2000		

Comments: Redshift = 0.9487

RA (degrees) = 148.628851114941, DEC (degrees) = 50.5389160396633
error_RA (arcsec) = 0.02475874275, error_DEC (arcsec) = 0.02090148393

Here I have provided the final target coordinates with errors. These are in the ICRS frame, and have been taken from the Tractor catalogues of Legacy Surveys DR10 and DR9. I have checked the position using the z-band images from Legacy Surveys. The astrometry of Legacy Surveys DR8 onwards is placed on the GAIA DR2 frame. The differences between the GAIA DR2 and GAIA DR3 frame are of the order of a few dozen micro-arcsec at most, and usually even much smaller. So I would think this would not matter for our case.
Category=Galaxy
Description=[Active galactic nuclei, Radio cores, Radio jets]
Extended=YES

#	Name	Target Coordinates	Targ. Coord. Corrections	Miscellaneous
(3)	J120002.30+572212.7	RA: 12 00 0.4131 (180.0017213d) Dec: +57 22 23.40 (57.37317d) Equinox: J2000		

Comments: Redshift = 1.254

RA (degrees) = 180.001721115672, DEC (degrees) = 57.3731678330445
error_RA (arcsec) = 0.07553277373, error_DEC (arcsec) = 0.07310620213

Here I have provided the final target coordinates with errors. These are in the ICRS frame, and have been taken from the Tractor catalogues of Legacy Surveys DR10 and DR9. I have checked the position using the z-band images from Legacy Surveys. The astrometry of Legacy Surveys DR8 onwards is placed on the GAIA DR2 frame. The differences between the GAIA DR2 and GAIA DR3 frame are of the order of a few dozen micro-arcsec at most, and usually even much smaller. So I would think this would not matter for our case.
Category=Galaxy
Description=[Active galactic nuclei, Radio cores, Radio jets]
Extended=YES

#	Name	Target Coordinates	Targ. Coord. Corrections	Miscellaneous
(4)	J153837.14+400640.2	RA: 15 38 37.3643 (234.6556846d) Dec: +40 06 36.40 (40.11011d) Equinox: J2000		

Comments: Redshift = 0.9754

RA (degrees) = 234.65568446903, DEC (degrees) = 40.1101110295052
error_RA (arcsec) = 0.01386840515, error_DEC (arcsec) = 0.01274685484

Here I have provided the final target coordinates with errors. These are in the ICRS frame, and have been taken from the Tractor catalogues of Legacy Surveys DR10 and DR9. I have checked the position using the z-band images from Legacy Surveys. The astrometry of Legacy Surveys DR8 onwards is placed on the GAIA DR2 frame. The differences between the GAIA DR2 and GAIA DR3 frame are of the order of a few dozen micro-arcsec at most, and usually even much smaller. So I would think this would not matter for our case.
Category=Galaxy
Description=[Active galactic nuclei, Radio cores, Radio jets]
Extended=YES

Observation

Proposal 9138, Observation 1: J012230.18+270832.3

Wed Jun 04 23:00:18 GMT 2025

Diagnostic Status: Warning

Observing Template: NIRSpec Fixed Slit Spectroscopy

Diagnostics

(Visit 1:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.

Fixed Targets

#	Name	Target Coordinates	Targ. Coord. Corrections	Miscellaneous
(1)	J012230.18+270832.3	RA: 01 22 29.9581 (20.6248254d) Dec: +27 08 39.78 (27.14438d) Equinox: J2000		

Comments: Redshift = 1.144

RA (degrees) = 20.6248256082825, DEC (degrees) = 27.1443821071447
error_RA (arcsec) = 0.05170400635, error_DEC (arcsec) = 0.05368708672

Here I have provided the final target coordinates with errors. These are in the ICRS frame, and have been taken from the Tractor catalogues of Legacy Surveys DR10 and DR9. I have checked the position using the z-band images from Legacy Surveys. The astrometry of Legacy Surveys DR8 onwards is placed on the GAIA DR2 frame. The differences between the GAIA DR2 and GAIA DR3 frame are of the order of a few dozen micro-arcsec at most, and usually even much smaller. So I would think this would not matter for our case.
Category=Galaxy
Description=[Active galaxies, Radio cores, Radio jets]
Extended=YES

Acquisition

#	Target	TA Method	Subarray	Filter	Readout Pattern	Groups/Int	Integrations/Exp	Total Integrations	Total Exposure Time	ETC Wkbk.Calc ID
1	SAME	WATA	FULL	CLEAR	NRSRAPID	3	1	1	42.947	223385.7

Template

HFF Readout Mode	Slit	Subarray
false	S400A1	FULL

Dithers

#	Primary Dither Positions	Sub-Pixel Pattern
1	3	SPATIAL

Spectral Elements

#	Grating/Filter	Slit	Readout Pattern	Groups/Int	Integrations/Exp	#	Autocal	Total Dithers	Total Integrations	Total Exposure Time	ETC Wkbk.Calc ID
1	G140M/F070LP	S400A1	NRSIRS2RAPID	40	2	1	NONE	6	12	7177.734	223385.4

Observation

Proposal 9138, Observation 2: J095431.30+503215.4

Wed Jun 04 23:00:18 GMT 2025

Diagnostic Status: Warning

Observing Template: NIRSpec Fixed Slit Spectroscopy

Diagnostics

(Visit 2:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.

Fixed Targets

#	Name	Target Coordinates	Targ. Coord. Corrections	Miscellaneous
(2)	J095431.30+503215.4	RA: 09 54 30.9243 (148.6288512d) Dec: +50 32 20.10 (50.53892d) Equinox: J2000		

Comments: Redshift = 0.9487

RA (degrees) = 148.628851114941, DEC (degrees) = 50.5389160396633
error_RA (arcsec) = 0.02475874275, error_DEC (arcsec) = 0.02090148393

Here I have provided the final target coordinates with errors. These are in the ICRS frame, and have been taken from the Tractor catalogues of Legacy Surveys DR10 and DR9. I have checked the position using the z-band images from Legacy Surveys. The astrometry of Legacy Surveys DR8 onwards is placed on the GAIA DR2 frame. The differences between the GAIA DR2 and GAIA DR3 frame are of the order of a few dozen micro-arcsec at most, and usually even much smaller. So I would think this would not matter for our case.
Category=Galaxy
Description=[Active galactic nuclei, Radio cores, Radio jets]
Extended=YES

Acquisition

#	Target	TA Method	Subarray	Filter	Readout Pattern	Groups/Int	Integrations/Exp	Total Integrations	Total Exposure Time	ETC Wkbk.Calc ID
1	SAME	WATA	FULL	CLEAR	NRSRAPID	3	1	1	42.947	223385.3

Template

HFF Readout Mode	Slit	Subarray
false	S400A1	FULL

Dithers

#	Primary Dither Positions	Sub-Pixel Pattern
1	3	SPATIAL

Spectral Elements

#	Grating/Filter	Slit	Readout Pattern	Groups/Int	Integrations/Exp	#	Autocal	Total Dithers	Total Integrations	Total Exposure Time	ETC Wkbk.Calc ID
1	G140M/F070LP	S400A1	NRSIRS2RAPID	40	2	1	NONE	6	12	7177.734	223385.6

Observation

Proposal 9138, Observation 3: J120002.30+572212.7

Wed Jun 04 23:00:18 GMT 2025

Diagnostic Status: Warning

Observing Template: NIRSpec Fixed Slit Spectroscopy

Diagnostics

(Visit 3:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.

Fixed Targets

#	Name	Target Coordinates	Targ. Coord. Corrections	Miscellaneous
(3)	J120002.30+572212.7	RA: 12 00 0.4131 (180.0017213d) Dec: +57 22 23.40 (57.37317d) Equinox: J2000		

Comments: Redshift = 1.254

RA (degrees) = 180.001721115672, DEC (degrees) = 57.3731678330445
error_RA (arcsec) = 0.07553277373, error_DEC (arcsec) = 0.07310620213

Here I have provided the final target coordinates with errors. These are in the ICRS frame, and have been taken from the Tractor catalogues of Legacy Surveys DR10 and DR9. I have checked the position using the z-band images from Legacy Surveys. The astrometry of Legacy Surveys DR8 onwards is placed on the GAIA DR2 frame. The differences between the GAIA DR2 and GAIA DR3 frame are of the order of a few dozen micro-arcsec at most, and usually even much smaller. So I would think this would not matter for our case.
Category=Galaxy
Description=[Active galactic nuclei, Radio cores, Radio jets]
Extended=YES

Acquisition

#	Target	TA Method	Subarray	Filter	Readout Pattern	Groups/Int	Integrations/Exp	Total Integrations	Total Exposure Time	ETC Wkbk.Calc ID
1	SAME	WATA	FULL	CLEAR	NRSRAPID	3	1	1	42.947	223385.8

Template

HFF Readout Mode	Slit	Subarray
false	S400A1	FULL

Dithers

#	Primary Dither Positions	Sub-Pixel Pattern
1	3	SPATIAL

Spectral Elements

#	Grating/Filter	Slit	Readout Pattern	Groups/Int	Integrations/Exp	#	Autocal	Total Dithers	Total Integrations	Total Exposure Time	ETC Wkbk.Calc ID
1	G140M/F070LP	S400A1	NRSIRS2RAPID	40	2	1	NONE	6	12	7177.734	223385.5

Observation

Proposal 9138, Observation 4: J153837.14+400640.2

Wed Jun 04 23:00:18 GMT 2025

Diagnostic Status: Warning

Observing Template: NIRSpec Fixed Slit Spectroscopy

Diagnostics

(Visit 4:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.

Fixed Targets

#	Name	Target Coordinates	Targ. Coord. Corrections	Miscellaneous
(4)	J153837.14+400640.2	RA: 15 38 37.3643 (234.6556846d) Dec: +40 06 36.40 (40.11011d) Equinox: J2000		

Comments: Redshift = 0.9754

RA (degrees) = 234.65568446903, DEC (degrees) = 40.1101110295052
error_RA (arcsec) = 0.01386840515, error_DEC (arcsec) = 0.01274685484

Here I have provided the final target coordinates with errors. These are in the ICRS frame, and have been taken from the Tractor catalogues of Legacy Surveys DR10 and DR9. I have checked the position using the z-band images from Legacy Surveys. The astrometry of Legacy Surveys DR8 onwards is placed on the GAIA DR2 frame. The differences between the GAIA DR2 and GAIA DR3 frame are of the order of a few dozen micro-arcsec at most, and usually even much smaller. So I would think this would not matter for our case.
Category=Galaxy
Description=[Active galactic nuclei, Radio cores, Radio jets]
Extended=YES

Acquisition

#	Target	TA Method	Subarray	Filter	Readout Pattern	Groups/Int	Integrations/Exp	Total Integrations	Total Exposure Time	ETC Wkbk.Calc ID
1	SAME	WATA	FULL	CLEAR	NRSRAPID	3	1	1	42.947	223385.9

Template

HFF Readout Mode	Slit	Subarray
false	S400A1	FULL

Dithers

#	Primary Dither Positions	Sub-Pixel Pattern
1	3	SPATIAL

Spectral Elements

#	Grating/Filter	Slit	Readout Pattern	Groups/Int	Integrations/Exp	#	Autocal	Total Dithers	Total Integrations	Total Exposure Time	ETC Wkbk.Calc ID
1	G140M/F070LP	S400A1	NRSIRS2RAPID	40	1	1	NONE	6	6	3588.867	223385.2